BUMRAH'S BOWLING BLUEPRINT

Among the most studied of bowlers, Bumrah remains an enigma to batsmen. The key to his unpredictability lies in his utterly unorthodox bowling action



RUN-UP EFFICIENCY

Neither particularly long nor especially fast, what it lacks in raw momentum is compensated through greater control, precision and optimal transfer of force. It eventually produces average speeds of around 140 kmph

BRACED KNEE

By locking the front knee, he converts the forward momentum of the run-up into a catapult effect. Instead of energy dissipating through a bent knee, it is driven upward through the body and into the bowling arm. The lower body abruptly halts while the upper body whips through

Illustration by **NILANJAN DAS / AI**

code"—derived from his superhuman ability to turn a match on its head whenever his captain so desires, video-game style—doesn't quite pass muster. For the simple reason that cheat codes can eventually be cracked; there's no mitigating the Bumrah hack.

Cracking the Bumrah Code

The primary reason behind that is his unorthodox action. Not only does it serve as deception, the physics behind it affords him the luxury of producing the extraordinary while using seemingly ordinary tricks. The first point of deception comes from a run-up that is not quite commensurate with the speeds he generates. Relatively short and neither rapid nor explosive in its approach to the crease, Bumrah is an anomaly in the 140 kmph club. But what it lacks in speed, his run-up compensates for by generating greater control and an optimum transfer of force into the ball.

It begins with his braced front leg at the point of delivery. By forcing the knee straight at release, the bowler transforms forward momentum into a catapult effect, pushing the energy from the run-up upward through the body and into the ball rather than letting it be absorbed (and partly wasted) by a bent knee. In effect, with that abrupt halt, it takes the lower-body momentum generated during the run-up and instantly transmits it up and



"ONCE-IN-A-GENERATION BOWLER, HE IS A NATIONAL TREASURE. HE KNOWS WHAT NEEDS TO BE DONE, HOW IT NEEDS TO BE DONE...THE BEST IN THE BUSINESS"

SURYAKUMAR YADAV
T20 Captain